



Science

Stage 9

Paper 1

2025

Cambridge Lower Secondary Progression Test

Mark Scheme

General guidelines on marking

Many descriptive answers can be expressed in a variety of ways. Professional judgement can be used in these cases, providing it matches the marking points and further information in the mark scheme.

Answers may have words spelt incorrectly. Credit is normally given for phonetically correct answers, unless the word has a scientifically different meaning. For example, where the answer should be antennae, credit will be given for antenna but not for anthen (too close to anther).

Only the science is being assessed, so answers do not need to be grammatically correct.

Significant figures will be indicated in the question or in the mark scheme.

Unless specified all marking points are independent.

Annotations and abbreviations

/ or	alternate responses for the same marking point
() brackets	the words or units in brackets do not need to be stated, for example, (recycles or releases or provides) minerals = minerals scores the mark
<u>underline</u>	exact word is required
Accept	an acceptable response
Do not accept	indicates an incorrect response that would contradict another otherwise correct alternative
Ignore	indicates an irrelevant answer that is not creditworthy. Full marks can still be achieved even with answers that are ignored.
Note	provides extra information when necessary
ecf	error carried forward; marks are awarded if an incorrect response has been carried forward from earlier working provided the subsequent working is correct
ora	or reverse argument; for example, as mass increases, volume increases could be written as mass decreases, volume decreases

Question	Answer	Marks	Further Information
1(a)	(prediction) decreases (explanation) less food for puffins or puffins starve	2	each correct answer = 1 mark Accept (idea that) puffins migrate to where the water is colder Ignore puffins die Accept population stays the same because the puffins find a new food source = 2 marks
1(b)	(population) decreases because predators eat puffins or decreases because predators eat eggs	1	

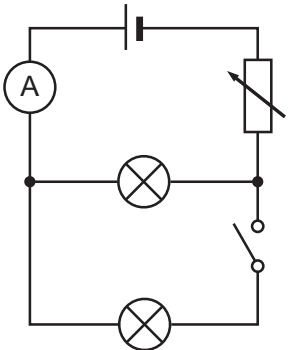
Question	Answer	Marks	Further Information
2(a)	E	1	Accept 2.8.8.2
2(b)	C	1	Accept 2.4
2(c)	A (and) B	1	both required in either order for the mark Accept 2.7 (and) 2.8.7
2(d)	A (and) C	1	both required in either order for the mark Accept 2.7 (and) 2.4

Question	Answer	Marks	Further Information
3	any three from (description) volume increases or sound gets louder pitch gets higher (explanation) amplitude increases frequency increases	3	each correct answer = 1 mark Note award marks wherever they are written Note assume answer refers to car getting closer to the object but Accept ora if car moving away from object is specified

Question	Answer	Marks	Further Information
4(a)	convection (currents)	1	
4(b)	any two from location of volcanoes location of earthquakes fossil records alignment of magnetic materials in the Earth's crust	2	each correct answer = 1 mark Note award marks wherever they are written
4(c)	the tectonic plates in the ocean are moving away from each other	1	Accept there are constructive plate boundaries in the ocean or there are divergent plate boundaries in the ocean Accept sea floor spreading

Question	Answer	Marks	Further Information
5(a)	(A) chromosome (B) DNA (C) gene	3	each correct answer = 1 mark Accept deoxyribonucleic acid
5(b)	gene	1	Accept C , but name takes precedence

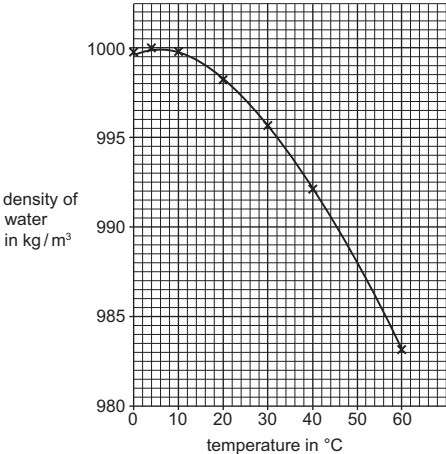
Question	Answer	Marks	Further Information
6(a)	lithium or francium	1	Accept Li or Fr, but name takes precedence
6(b)	any number in the range 680 to 740 ($^{\circ}\text{C}$)	1	
6(c)	(idea that) increases	1	

Question	Answer	Marks	Further Information
7(a)	2 (A) 2 (A)	1	both correct for the mark
7(b)		3	correct symbol for variable resistor = 1 mark correct symbol for ammeter and open or closed switch = 1 mark wires connected correctly and no gaps in the circuit by eye = 1 mark Ignore junction of connectors symbols Ignore any labels or words

Question	Answer	Marks	Further Information
8	respiration decomposition	2	each correct answer = 1 mark Do not accept breathing or photosynthesis Accept decay

Question	Answer	Marks	Further Information
9	(dark mice) decreases and (light mice decreases and then) increases (explanation) (idea that) dark mice are more visible so are more likely to be eaten or light mice are less visible so less likely to be eaten (idea that) light mice are more likely to reproduce or dark mice are less likely to reproduce (idea that) the characteristics for light mice more likely to be passed on to offspring or characteristics for dark mice less likely to be passed on to offspring	4	each correct answer = 1 mark Accept dark mice goes from 4 to 1 and light mice goes from 4 to (3 to) 7

Question	Answer	Marks	Further Information
10	(idea that) Blessy adds excess zinc carbonate to dilute sulfuric acid or AW filters the mixture slowly evaporates (the filtrate) or boils away half of filtrate or leaves (filtrate) to crystallise slowly	3	each correct answer = 1 mark Accept filtration Ignore sieving Do not accept heat to dryness or boil to dryness Do not accept evaporate residue slowly

Question	Answer	Marks	Further Information
11(a)		2	all points plotted correctly = 1 mark Accept plots $\pm$ half a small square smooth curve through plotted points = 1 mark Accept ecf for the curve from incorrect plots Do not accept dot-to-dot or a straight line
11(b)	float (no marks) and (idea that) ice has a lower density than water or water has a higher density than ice water has a density of $999.8 \text{ (kg/m}^3\text{)}$ or the difference in density is $999.8 - 918 \text{ (kg/m}^3\text{)}$ or the difference in density is $81.8 \text{ (kg/m}^3\text{)}$	2	each correct answer = 1 mark Note full marks can only be awarded if 'float' is stated

Question	Answer	Marks	Further Information
12(a)	C A D B C (idea that) small planet is close to the Earth or small planet moves towards the Earth or large rock moves towards Earth A (idea that) small planet collides with the Earth or there is a collision D (idea that) debris collects around the Earth or dust is formed around the Earth B (idea that) the dust comes together to form the Moon	3	correct order = 1 mark all four descriptions matching the diagrams = 2 marks for the descriptions only two or three descriptions matching the diagrams = 1 mark for the descriptions one description matching the diagrams = 0 marks for the descriptions
12(b)	any one from rock samples from the Moon are similar to rock on Earth lunar meteorites are similar in composition to the Earth	1	Accept any suitable higher-level evidence, e.g. Earth's spin and the Moon's orbit have similar orientations or there here is evidence in other star systems of similar collisions

Question	Answer	Marks	Further Information
13	(strength) (idea that) the mechanism of cleaning is similar, e.g. the useful substances are separated and then returned or (idea that) it visualises a process that cannot be seen inside the kidney (limitation) (idea that) not all substances are removed by the kidney from the blood, but all the contents are removed from the bag or (idea that) the filtering happens within the kidney, but everything is taken out of the bag for cleaning	2	correct strength = 1 mark correct limitation = 1 mark Accept named examples, e.g. red blood cells are not removed by the kidney

Question	Answer	Marks	Further Information
14(a)	(I think) that the smaller the pieces of wood the faster they will burn (because) smaller pieces of wood have a larger surface area.	2	each correct answer = 1 mark Accept reverse argument for both marking points
14(b)	any two from mass of wood sample time for wood to burn completely surface area of wood sample	2	each correct answer = 1 mark Accept volume of wood sample
Question	Answer	Marks	Further Information
15(a)	(idea that) level of liquid in thermometers is the same	1	
15(b)	(idea that for the same temperature) the volumes of water are different	1	Accept the height of the water is different

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